

MOUNT KENYA UNIVERSITY
SCHOOL OF EDUCATION
DEPARTMENT OF CURRICULUM AND INSTRUCTION

**TEACHERS' COMPETENCE AND ITS INFLUENCE ON THE EFFECTIVE
IMPLEMENTATION OF THE COMPETENCY-BASED CURRICULUM IN
PUBLIC PRIMARY SCHOOLS IN HABASWEIN SUB-COUNTY, WAJIR
COUNTY, KENYA**

[FULL NAME OF STUDENT]

[REGISTRATION NUMBER]

A Research Proposal Submitted to the School of Education in Partial Fulfilment of the
Requirements for the Award of the Degree of Master of Education (Curriculum
Studies) of Mount Kenya University

SEPTEMBER 2026

DECLARATION

This research proposal is my original work and has not been presented for the award of a degree or for any other award in any other university.

Signature: _____ Date: _____

[Full Name of Student]

[Registration Number]

Supervisors' Approval

This research proposal has been submitted for examination with our approval as the university supervisors.

Signature: _____ Date: _____

[Name of Supervisor 1]

Department of Curriculum and Instruction, School of Education, Mount Kenya University

Signature: _____ Date: _____

[Name of Supervisor 2]

Department of Curriculum and Instruction, School of Education, Mount Kenya University

DEDICATION

I dedicate this work to my family, whose patience, prayers and encouragement have carried me through my postgraduate studies, and to the teachers of Habaswein Sub-County, who continue to serve their learners with dedication under some of the most demanding conditions in the country.

ACKNOWLEDGEMENT

I thank the Almighty God for the gift of life, good health and the strength to undertake this work. I am deeply grateful to my supervisors for their patient guidance, constructive criticism and encouragement at every stage of developing this proposal. Their insights have greatly shaped the direction and quality of the study.

I also thank the lecturers and staff of the School of Education at Mount Kenya University for the knowledge and support I have received throughout the programme. I appreciate my classmates for the many discussions that sharpened my thinking, and my colleagues in the teaching profession for sharing their experiences of implementing the Competency-Based Curriculum. Finally, I am indebted to my family for their understanding and sacrifice during the many hours I spent away from them while working on this proposal. May God bless you all.

TABLE OF CONTENTS

DECLARATION.....	ii
DEDICATION.....	iii
ACKNOWLEDGEMENT.....	iv
TABLE OF CONTENTS.....	v
LIST OF TABLES.....	vii
LIST OF FIGURES.....	viii
ABBREVIATIONS AND ACRONYMS.....	ix
ABSTRACT.....	x
CHAPTER ONE: INTRODUCTION.....	1
1.1 Background to the Study.....	1
1.2 Statement of the Problem.....	3
1.3 Purpose of the Study.....	4
1.4 Research Objectives.....	5
1.4.1 General Objective.....	5
1.4.2 Specific Objectives.....	5
1.5 Research Questions.....	5
1.6 Research Hypotheses.....	6
1.7 Justification of the Study.....	6
1.8 Significance of the Study.....	6
1.9 Limitations of the Study.....	7
1.10 Delimitations of the Study.....	7
1.11 Assumptions of the Study.....	8
1.12 Operational Definition of Terms.....	8
1.13 Organisation of the Proposal.....	9
CHAPTER TWO: LITERATURE REVIEW.....	10
2.1 Introduction.....	10
2.2 Theoretical Review.....	10
2.2.1 Concerns-Based Adoption Model.....	10
2.2.2 Teacher Professional Competence Model.....	11
2.2.3 Technological Pedagogical Content Knowledge Framework.....	12
2.3 The Competency-Based Curriculum in Kenya.....	13
2.4 Empirical Review.....	14
2.4.1 Teachers’ Knowledge and Understanding of the Competency-Based Curriculum.....	14
2.4.2 Teachers’ Pedagogical Competence in Implementing the CBC.....	15
2.4.3 Factors Affecting Teachers’ Competence in Implementing the CBC.....	16
2.4.4 Teachers’ Competence and Effective Implementation of the CBC.....	18
2.5 Critique of Existing Literature.....	19
2.6 Conceptual Framework.....	20

2.7 Summary of Literature and Research Gaps.....	22
CHAPTER THREE: RESEARCH METHODOLOGY.....	24
3.1 Introduction.....	24
3.2 Research Design.....	24
3.3 Location of the Study.....	24
3.4 Target Population.....	25
3.5 Sampling Techniques and Sample Size.....	26
3.5.1 Sampling Techniques.....	26
3.5.2 Sample Size.....	27
3.6 Research Instruments.....	28
3.6.1 Questionnaire for Teachers.....	28
3.6.2 Interview Guide for Head Teachers and Curriculum Support Officers. .	28
3.6.3 Classroom Observation Checklist.....	29
3.6.4 Operationalisation of Study Variables.....	29
3.7 Pilot Study.....	30
3.7.1 Validity of the Research Instruments.....	30
3.7.2 Reliability of the Research Instruments.....	30
3.7.3 Trustworthiness of Qualitative Data.....	31
3.8 Data Collection Procedures.....	31
3.9 Data Analysis and Presentation.....	32
3.9.1 Quantitative Data Analysis.....	32
3.9.2 Qualitative Data Analysis.....	32
3.9.3 Integration of Quantitative and Qualitative Findings.....	33
3.10 Logistical Considerations.....	33
3.11 Ethical Considerations.....	33
REFERENCES.....	35
APPENDICES.....	39
APPENDIX I: LETTER OF INTRODUCTION.....	40
APPENDIX II: INFORMED CONSENT FORM.....	41
APPENDIX III: QUESTIONNAIRE FOR TEACHERS.....	42
Section A: Demographic Information.....	42
Section B: Teachers’ Knowledge and Understanding of the CBC.....	42
Section C: Teachers’ Pedagogical Competence.....	43
Section D: Factors Affecting Teachers’ Competence in Implementing the CBC.....	44
Section E: Effective Implementation of the CBC.....	45
Section F: General Comments.....	45
APPENDIX IV: INTERVIEW GUIDE FOR HEAD TEACHERS AND CURRICULUM SUPPORT OFFICERS.....	46
APPENDIX V: CLASSROOM OBSERVATION CHECKLIST.....	48
APPENDIX VI: WORK PLAN.....	49
APPENDIX VII: RESEARCH BUDGET.....	50

LIST OF TABLES

Table 2.1: Summary of Research Gaps.....	22
Table 3.1: Target Population.....	26
Table 3.2: Sample Size Distribution.....	27
Table 3.3: Operationalisation of Study Variables.....	29
Table A.1: Work Plan, October 2026 – September 2027.....	49
Table A.2: Research Budget.....	50

LIST OF FIGURES

Figure 2.1: Conceptual Framework.....	21
---------------------------------------	----

ABBREVIATIONS AND ACRONYMS

APHRC	African Population and Health Research Center
ASAL	Arid and Semi-Arid Lands
BOM	Board of Management
CBA	Competency-Based Assessment
CBAM	Concerns-Based Adoption Model
CBC	Competency-Based Curriculum
CBE	Competency-Based Education
CIDP	County Integrated Development Plan
CPD	Continuous Professional Development
CSO	Curriculum Support Officer
CVI	Content Validity Index
ICT	Information and Communication Technology
KICD	Kenya Institute of Curriculum Development
KNEC	Kenya National Examinations Council
MKU	Mount Kenya University
MoE	Ministry of Education
NACOSTI	National Commission for Science, Technology and Innovation
PCIs	Pertinent and Contemporary Issues
PWPER	Presidential Working Party on Education Reform
SPSS	Statistical Package for the Social Sciences
TPACK	Technological Pedagogical Content Knowledge
TSC	Teachers Service Commission
UNESCO	United Nations Educational, Scientific and Cultural Organization

ABSTRACT

The Competency-Based Curriculum (CBC) introduced in Kenya in 2017 seeks to develop learners who can think critically, solve problems, communicate and apply knowledge in real life. Its success depends largely on teachers, who are expected to understand the philosophy of the curriculum, use learner-centred pedagogy and assess learners through a range of authentic tools. Evidence from different parts of the country, however, suggests that many teachers have not yet acquired the competence the curriculum demands, and that this gap may be wider in marginalised arid and semi-arid counties such as Wajir. The purpose of this study is to assess teachers' competence and its influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County, Wajir County, Kenya. The study is guided by four objectives: to assess the level of teachers' knowledge and understanding of the principles and practices of the CBC; to analyse the level of teachers' pedagogical competence in implementing the CBC; to identify the factors affecting teachers' competence; and to establish the relationship between teachers' competence and the effective implementation of the CBC. The study is anchored on the Concerns-Based Adoption Model, the Teacher Professional Competence Model and the Technological Pedagogical Content Knowledge framework. A convergent parallel mixed methods design combining a descriptive survey and a correlational approach will be used. The target population comprises approximately 42 public primary schools, 42 head teachers, 336 teachers and 3 Curriculum Support Officers. Stratified random sampling will be used to select 21 schools, and a sample of 183 teachers will be determined using a finite population formula and distributed proportionately across the strata. The 21 head teachers of the sampled schools and all 3 Curriculum Support Officers will also take part. Data will be collected through questionnaires, semi-structured interviews and classroom observation checklists. The instruments will be piloted in Wajir South Sub-County, content validity will be established through expert judgement and reliability will be tested using Cronbach's alpha. Quantitative data will be analysed using descriptive statistics, Pearson correlation and multiple linear regression with the aid of SPSS, while qualitative data will be analysed thematically. The findings are expected to inform the Ministry of Education, the Teachers Service Commission, the Kenya Institute of Curriculum Development and the County Government of Wajir in designing targeted teacher support for marginalised areas.

Keywords: Teachers' competence, Competency-Based Curriculum, curriculum implementation, pedagogical competence, marginalised areas, Wajir County

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Across the world, school systems are steadily moving away from curricula that reward the memorisation of content towards curricula that ask learners to do something meaningful with what they know. This shift, commonly described as competency-based education, places the emphasis on skills, values and attitudes that learners can apply in real situations, rather than on the volume of facts they can reproduce in an examination (Jentsch & König, 2022). The global push for this kind of learning has come at the same time as a serious shortage of qualified teachers. UNESCO and the International Task Force on Teachers for Education 2030 (2024) estimate that the world will need 44 million additional primary and secondary teachers by 2030, with sub-Saharan Africa alone requiring about 15 million. The report stresses that recruiting more teachers is not enough; the teachers already in classrooms must be supported to acquire the professional competence that new curricula demand.

Competence, in the professional sense, is more than a teaching certificate. Jentsch and König (2022) describe teacher competence as a combination of professional knowledge, practical skills, beliefs and motivation that together allow a teacher to cope with the demands of real classrooms. Their review of international large-scale studies shows that these cognitive and affective dimensions develop over time, both during initial training and through continuous professional development, and that they are closely linked to the quality of instruction learners receive. Mishra et al. (2023) add that the knowledge base of teachers is not static; as curricula and technologies evolve, teachers must continually integrate content knowledge, pedagogical knowledge and technological knowledge if their teaching is to remain relevant. In short, a curriculum can only be as good as the teachers who deliver it.

On the African continent, several countries have adopted competence-based curricula in the past decade, and their experiences offer useful lessons. Wawire et al. (2026), in a scoping review covering Ghana, Kenya, Rwanda, South Africa, Tanzania, Uganda and Zambia, found that although policy documents across the region promote learner-centred pedagogy and formative assessment, classroom practice in many schools still leans towards teacher talk, copying of notes and summative testing. In Tanzania, Kasuga and Kalolo (2025) documented a set of dilemmas in Njombe Region, among them inadequate teacher preparation, an uneven distribution of teaching resources and a tension between the new competency approach and societal expectations that remain tied to examination results. Uganda's experience with its revised lower secondary

curriculum tells a similar story. Kidega et al. (2024) reported that teachers in Gulu City struggled with large classes, limited materials and insufficient training, while Barasa et al. (2025) raised concerns that schools serving poorer communities were falling behind in implementing the reform, thereby widening rather than narrowing inequality.

Kenya began its own journey in 2017, when the Competency-Based Curriculum (CBC) replaced the 8-4-4 system in a phased rollout that started with pre-primary and lower primary grades. The reform was designed to nurture every learner's potential through seven core competencies: communication and collaboration, critical thinking and problem solving, imagination and creativity, citizenship, digital literacy, learning to learn, and self-efficacy (Akala, 2021). It introduced a new structure in which learners spend two years in pre-primary, six years in primary, three years in junior school and three years in senior school before proceeding to tertiary education. In 2022, the government constituted the Presidential Working Party on Education Reform (PWPER) to review the implementation of the curriculum. The Working Party endorsed the competency approach but recommended a reduction in the number of learning areas, a lighter assessment load and a stronger programme of teacher capacity building (Presidential Working Party on Education Reform [PWPER], 2023). Following these recommendations, the reform has increasingly been referred to as Competency-Based Education (CBE), although the classroom expectations placed on teachers remain largely the same.

A central feature of the Kenyan reform is its assessment model. The Kenya National Examinations Council (KNEC, 2021) developed a competency-based assessment framework that requires teachers to use a range of formative tools such as observation schedules, rating scales, checklists, portfolios, projects and rubrics, alongside summative assessment. This model assumes that teachers can design authentic tasks, judge learner performance against clear criteria and keep accurate records for every child. Macheso, Kati, and Wafula (2024) found that, even several years into the reform, many primary school teachers in western Kenya were still relying on a narrow range of assessment tools, with portfolios, anecdotal records and projects being used far less often than written tests. Such findings suggest that the competence required by the curriculum has not been fully acquired by the teachers expected to implement it.

The wider body of Kenyan research points in the same direction. Studies conducted in Nairobi, Nyamira, Kirinyaga, Kinangop, Moiben and Bomet have repeatedly found that teachers were either inadequately trained for the new curriculum or trained in a hurried manner that left gaps in their understanding (Isaboke et al., 2021; Karua et al., 2025; Keter & Wabuke, 2025; Mohamed et al., 2022; Mutai et al., 2024; Okeyo & Mokua, 2023). Mwangi et al. (2026) observed that although 95% of teachers in their sample had

attended CBC training, 65% still held negative perceptions of the curriculum, which suggests that attendance at a workshop does not automatically translate into competence or commitment. Similarly, Mwang'ombe (2021) noted that teachers' understanding of CBC was often vague, and that this vagueness showed up in lesson planning, classroom delivery and assessment. In a large classroom-based study, the African Population and Health Research Center (APHRC, 2022) found that while some teachers delivered effective CBC lessons, many lessons fell short on learner engagement and the integration of core competencies.

The difficulties described above are likely to be more pronounced in the arid and semi-arid lands (ASAL) of northern Kenya. Hussein (2023) reported that poor infrastructure, a shortage of learning resources and limited teacher training undermined CBC implementation across northern Kenya, while Ongubo (2025) found that more than 60% of public primary school teachers in Township Ward, Garissa County, had not received comprehensive in-service training on CBC. Wajir County, where this study is located, shares many of these characteristics. The County Integrated Development Plan acknowledges persistent challenges in the education sector, including a shortage of teachers, long distances to school, the effects of recurrent drought on enrolment and attendance, and a pastoralist lifestyle that disrupts continuous schooling (County Government of Wajir, 2023). Insecurity along the border with Somalia has also led to the transfer of many non-local teachers, leaving schools understaffed and heavily reliant on teachers employed by Boards of Management.

Habaswein Sub-County lies in the southern part of Wajir County, along the Garissa to Wajir road. Most of its public primary schools serve pastoralist and agro-pastoralist communities, and a number of them operate as low-cost boarding schools to keep children from mobile households in school. Teachers in these schools work under demanding conditions: classes can be large, instructional materials are few, electricity and internet access are unreliable, and opportunities for professional development are scarce. Yet the curriculum they are expected to deliver is exactly the same as the one taught in well-resourced urban schools, and their learners sit the same national assessments. It is therefore important to understand how competent teachers in Habaswein are to implement the CBC, what shapes that competence, and how it relates to the quality of implementation in their classrooms.

1.2 Statement of the Problem

The success of the CBC depends on teachers who understand its philosophy, use learner-centred methods, integrate core competencies and pertinent and contemporary issues into their lessons, and assess learners in authentic ways. Ideally, every public

primary school teacher in Kenya should possess these competences, and the classroom experience of a learner in Habaswein should be comparable to that of a learner in Nairobi or Nakuru. The government has made deliberate efforts towards this ideal through national training programmes, the supply of curriculum designs and textbooks, and the work of Curriculum Support Officers (PWPER, 2023).

In reality, the evidence suggests a wide gap between what the curriculum expects of teachers and what many teachers are able to do. Studies across the country show that a considerable proportion of teachers have limited understanding of CBC principles, find it difficult to prepare schemes of work and lesson plans in the required format, and rely on a narrow set of assessment tools (Karua et al., 2025; Macheso, Kati, & Wafula, 2024; Mwang'ombe, 2021). In northern Kenya, these gaps are compounded by teacher shortages, insecurity, inadequate infrastructure and the limited reach of in-service training (Hussein, 2023; Ongubo, 2025). When teachers lack the necessary competence, the likely consequence is superficial implementation, in which the language of the new curriculum is adopted while classroom practice remains largely teacher-centred and examination-driven (Awiti, 2023; Wawire et al., 2026). Learners in marginalised areas then miss out on the very competencies that the reform was meant to develop, and existing inequalities are deepened rather than reduced (Barasa et al., 2025).

Although many studies have examined teacher preparedness for CBC, most of them were conducted in urban or relatively well-served counties such as Nairobi, Machakos, Kirinyaga, Nyandarua and Uasin Gishu (Isaboke et al., 2021; Mogere, 2022; Mutua & Waweru, 2023). Very little empirical work has been done in Wajir County, and none that the researcher is aware of has examined teachers' competence and its influence on CBC implementation in Habaswein Sub-County. Moreover, many previous studies measured preparedness mainly in terms of attendance at training, without assessing teachers' actual knowledge, pedagogical practice and assessment competence, or statistically testing how these dimensions relate to the effectiveness of implementation. This lack of context-specific evidence makes it difficult for the Ministry of Education, the Teachers Service Commission and the County Government of Wajir to design targeted interventions. The present study therefore seeks to assess teachers' competence and establish its influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County, Wajir County.

1.3 Purpose of the Study

The purpose of this study is to assess the competence of teachers and to establish how that competence influences the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County, Wajir County, Kenya,

with a view to generating evidence that can inform teacher support and policy in marginalised settings.

1.4 Research Objectives

1.4.1 General Objective

The general objective of the study is to assess teachers' competence and its influence on the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County, Wajir County, Kenya.

1.4.2 Specific Objectives

The study will be guided by the following specific objectives:

- i. To assess the level of teachers' knowledge and understanding of the principles and practices of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- ii. To analyse the level of teachers' pedagogical competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- iii. To identify the factors affecting teachers' competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- iv. To establish the relationship between teachers' competence and the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.

1.5 Research Questions

The study will seek to answer the following research questions:

- i. What is the level of teachers' knowledge and understanding of the principles and practices of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- ii. What is the level of teachers' pedagogical competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- iii. What factors affect teachers' competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?

- iv. What is the relationship between teachers' competence and the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?

1.6 Research Hypotheses

To give statistical direction to the fourth objective, the following null hypotheses will be tested at the 0.05 level of significance:

- H01: There is no statistically significant relationship between teachers' knowledge and understanding of the CBC and the effective implementation of the CBC in public primary schools in Habaswein Sub-County.
- H02: There is no statistically significant relationship between teachers' pedagogical competence and the effective implementation of the CBC in public primary schools in Habaswein Sub-County.
- H03: Teachers' overall competence has no statistically significant influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County.

1.7 Justification of the Study

Several reasons justify this study. First, the CBC is one of the most far-reaching education reforms in Kenya's history, and its first full cohort of learners has only recently transitioned into junior and senior school. Evidence on how well it is being implemented in different parts of the country is therefore both timely and necessary. Second, the reform was meant to promote equity, yet studies from other countries in the region show that competence-based reforms can widen gaps between advantaged and disadvantaged schools when teacher support is uneven (Barasa et al., 2025; Kasuga & Kalolo, 2025). Habaswein represents precisely the kind of marginalised setting in which such gaps are likely to appear. Third, the PWPER (2023) placed strong emphasis on teacher capacity building, and the Teachers Service Commission continues to invest in in-service training. For these investments to be effective, they must be informed by evidence on where teachers' competence is strong, where it is weak and what factors hold it back. Finally, most of the existing Kenyan literature is concentrated in central, western and urban regions. By focusing on a sub-county in the North Eastern region, this study will help to balance the national evidence base.

1.8 Significance of the Study

The findings of the study are expected to benefit several groups. The Ministry of Education and the Kenya Institute of Curriculum Development (KICD) may use the

findings to refine curriculum support strategies for ASAL counties, including the design of simplified curriculum materials and context-sensitive guidance on assessment. The Teachers Service Commission (TSC) may draw on the results to design targeted in-service training and teacher professional development programmes, and to reconsider staffing and incentive policies for hardship areas.

Teacher training colleges and universities may use the findings to strengthen the pre-service preparation of teachers, especially in learner-centred pedagogy, competency-based assessment and teaching in multi-grade and resource-constrained classrooms. The County Government of Wajir, through its education department, may find the results useful in planning early childhood and basic education support, as well as in lobbying for additional national resources. Head teachers and Curriculum Support Officers in Habaswein may use the findings to plan school-based professional learning, peer mentoring and instructional supervision. Teachers themselves may gain insight into the specific areas of their practice that need strengthening. Finally, researchers and scholars will have an additional reference on CBC implementation in marginalised contexts, and the study may open avenues for further research.

1.9 Limitations of the Study

The study is likely to face certain limitations. First, part of the data will be self-reported by teachers through questionnaires, and some respondents may overstate their competence to present themselves favourably. To reduce this risk, the researcher will triangulate questionnaire responses with classroom observations and interviews with head teachers and Curriculum Support Officers. Second, the vast distances between schools, poor road networks and occasional insecurity in parts of the sub-county may make some schools difficult to reach. The researcher will plan fieldwork during the dry season, coordinate with local education and security offices, and use trained research assistants familiar with the area. Third, some schools may be temporarily closed or have reduced attendance because of drought or the movement of pastoralist families. Where this happens, replacement schools will be drawn from the same stratum. Fourth, because the study is limited to one sub-county, its findings may not be fully generalisable to other parts of Kenya, although they are expected to be relevant to other ASAL settings with similar characteristics.

1.10 Delimitations of the Study

The study will be confined to public primary schools in Habaswein Sub-County, Wajir County. Private primary schools, pre-primary centres and junior schools will be excluded because their staffing, resourcing and management differ in ways that would complicate comparison. The respondents will be limited to TSC-employed and Board

of Management teachers who teach in Grades 1 to 6, head teachers of the sampled schools and Curriculum Support Officers serving the sub-county. In terms of content, the study will focus on four dimensions: teachers' knowledge and understanding of CBC, their pedagogical competence, the factors that affect their competence, and the relationship between competence and effective implementation. Other factors, such as parental involvement and learner characteristics, will be acknowledged but not examined in depth.

1.11 Assumptions of the Study

The study will be based on the following assumptions:

- i. That respondents will give honest and accurate information about their knowledge, practices and experiences in implementing the CBC.
- ii. That all public primary schools in Habaswein Sub-County are implementing the CBC in line with national guidelines issued by the Ministry of Education and KICD.
- iii. That teachers in the sampled schools have had at least some exposure to CBC training, whether through national rollout programmes, school-based sessions or self-directed learning.
- iv. That the security situation in the sub-county will remain stable enough to allow data collection within the planned period.

1.12 Operational Definition of Terms

Competency-Based Curriculum (CBC): The curriculum introduced in Kenya from 2017 that focuses on developing learners' core competencies, values and practical skills through learner-centred teaching and continuous assessment. In this study, the term also covers the current reference to Competency-Based Education (CBE).

Teachers' competence: The combination of knowledge, pedagogical skills, assessment skills and attitudes that enable a teacher to plan, deliver and assess CBC lessons as intended. In this study it will be measured through teachers' knowledge of CBC, their pedagogical competence and their assessment practice.

Knowledge and understanding of CBC: The extent to which a teacher can correctly explain the goals, core competencies, values, pertinent and contemporary issues, learning areas and assessment requirements of the CBC.

Pedagogical competence: A teacher's ability to plan and deliver lessons using learner-centred methods, integrate core competencies and values, manage learner activities, and use locally available resources effectively.

Effective implementation of CBC: The degree to which classroom practice reflects the intentions of the CBC, measured by learner engagement, integration of core competencies, use of competency-based assessment, lesson planning in the prescribed format and timely coverage of learning areas.

Factors affecting competence: School-based, teacher-based and contextual conditions, such as training, workload, resources, supervision and the physical environment, that enhance or limit a teacher's ability to implement the CBC.

Marginalised area: A region that experiences low access to social services, poor infrastructure and limited economic opportunity, as is the case in many ASAL counties including Wajir.

1.13 Organisation of the Proposal

This proposal is organised into three chapters. Chapter One presents the background to the study, the statement of the problem, the purpose, objectives, research questions and hypotheses, justification, significance, limitations, delimitations, assumptions and operational definitions of key terms. Chapter Two reviews the theories that underpin the study, examines empirical literature related to each objective, presents the conceptual framework and identifies the research gaps. Chapter Three describes the research methodology, including the research design, location, target population, sampling procedures, research instruments, piloting, data collection and analysis procedures, and ethical considerations. The references and appendices follow.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the literature that informs the study. It begins with the theories that explain how teachers take up and implement curriculum innovations and what constitutes teacher competence. It then describes the nature of the Competency-Based Curriculum in Kenya before reviewing empirical studies under each of the four study objectives. The chapter closes with a critique of the existing literature, the conceptual framework that links the study variables, and a summary of the research gaps the study intends to fill.

2.2 Theoretical Review

Three theories provide the lens for this study. The Concerns-Based Adoption Model explains how teachers move through stages of concern and levels of use when a new curriculum is introduced. The Teacher Professional Competence Model explains what teacher competence consists of and how it translates into classroom performance. Finally, the Technological Pedagogical Content Knowledge framework explains the integrated knowledge that teachers need to teach specific content well, including through the use of technology, which is one of the core competencies in the CBC.

2.2.1 Concerns-Based Adoption Model

The Concerns-Based Adoption Model (CBAM) was developed by Gene Hall and his colleagues at the University of Texas in the 1970s as a way of understanding how individuals, especially teachers, experience and adopt educational innovations. The model rests on the idea that change is a process rather than an event, and that it is carried out by individuals before it becomes an institutional reality. It has three diagnostic dimensions: stages of concern, levels of use and innovation configurations (Okeyo & Mokua, 2023).

The stages of concern describe how a teacher's feelings and worries about an innovation evolve. At the earliest stages, teachers are mainly concerned with getting information about the innovation and understanding how it will affect them personally. As they begin using it, their concerns shift to management issues such as time, materials and organisation. At the more advanced stages, their concerns focus on the consequences of the innovation for learners, on collaboration with colleagues and on refining the innovation to make it work better. The levels of use describe what teachers actually do with the innovation, ranging from non-use, through mechanical and routine use, to

refinement and renewal. Innovation configurations, on the other hand, describe the different forms an innovation takes when it is adapted in practice by different teachers (Okeyo & Mokua, 2023).

CBAM has been widely applied in African curriculum reform studies. In Kenya, Okeyo and Mokua (2023) used it to examine teachers' preparedness for CBC in Nyamira South and found that most teachers were still at early stages of concern, anxious about what the curriculum required of them. The model is relevant to the present study because it offers a way of interpreting teachers' knowledge and practice as points on a developmental path. A teacher who has only superficial knowledge of CBC is likely to be at an informational or personal stage of concern and a mechanical level of use, while a highly competent teacher is likely to be concerned with learner outcomes and to refine the curriculum creatively. The model also highlights that teachers' concerns must be addressed through appropriate support, which links directly to the third objective on factors affecting competence.

The main criticism of CBAM is that it focuses on the individual teacher and pays relatively little attention to structural conditions such as funding, class size, infrastructure and policy coherence, which may block implementation regardless of the teacher's stage of concern. It has also been criticised for suggesting a neat, linear progression, whereas teachers may move back and forth between stages when conditions change. For these reasons, the present study complements CBAM with a model of teacher competence and pays explicit attention to contextual factors in Habaswein.

2.2.2 Teacher Professional Competence Model

The second theory is the Teacher Professional Competence Model, which has its roots in the work of German and international researchers, notably Sigrid Blömeke and colleagues, who proposed that competence should be understood as a continuum. According to this model, teachers' dispositions, which include professional knowledge as well as beliefs, motivation and self-regulation, feed into situation-specific skills: the ability to perceive what is happening in a classroom, interpret it and decide how to respond. These skills, in turn, produce observable performance in the classroom (Jentsch & König, 2022).

Jentsch and König (2022) explain that the cognitive part of competence includes content knowledge, pedagogical content knowledge and general pedagogical knowledge, while the affective-motivational part includes teachers' beliefs about teaching and learning, their enthusiasm and their self-efficacy. They show, drawing on international large-scale studies, that both parts matter for the quality of instruction and

for learner achievement, and that both can be developed through initial teacher education and continuous professional development. An important implication of the model is that knowledge on its own does not guarantee good teaching. A teacher may know the principles of the CBC but still fail to apply them if they lack confidence, hold negative beliefs about the curriculum or cannot read and respond to classroom situations.

This model is useful for the present study for three reasons. First, it offers a clear structure for the independent variable, allowing teachers' competence to be broken down into knowledge and understanding of CBC (dispositions) and pedagogical competence (situation-specific skills and performance). Second, it explains why training attendance alone is a weak indicator of competence, which is a key criticism of earlier CBC studies. Third, it suggests that competence can be observed in practice, which justifies the use of classroom observation alongside questionnaires. The model's limitation is that it was developed largely in well-resourced systems and may underestimate the extent to which contextual constraints in marginalised areas limit what competent teachers can do. The study addresses this by examining contextual factors under the third objective.

2.2.3 Technological Pedagogical Content Knowledge Framework

The third lens is the Technological Pedagogical Content Knowledge (TPACK) framework, first proposed by Punya Mishra and Matthew Koehler in the mid-2000s as an extension of the notion of pedagogical content knowledge. The framework holds that effective teaching requires an integration of three kinds of knowledge: content knowledge (what to teach), pedagogical knowledge (how to teach) and technological knowledge (the tools that can support teaching). The areas where these types of knowledge overlap, such as pedagogical content knowledge and technological pedagogical knowledge, represent the specialised knowledge that distinguishes an expert teacher from a subject specialist or a technology enthusiast (Mishra et al., 2023).

Mishra et al. (2023) revisited the framework in light of new technologies and argued that teachers must also consider contextual knowledge, which they describe as the knowledge of the circumstances, constraints and opportunities in which teaching takes place. This addition is highly relevant to Habaswein, where the availability of electricity, devices and internet connectivity cannot be taken for granted. Digital literacy is one of the core competencies in the CBC, and teachers are expected to integrate it across learning areas. Murithi and Yoo (2021) found that many Kenyan primary teachers lacked the capacity to integrate technology meaningfully into CBC lessons, having received only basic computer literacy training. TPACK therefore helps

the study to examine whether teachers can combine content, pedagogy and available technology, including low-cost resources, in ways that promote the intended competencies. Its main limitation is that its knowledge domains overlap and can be difficult to measure separately; the study will therefore use it as an interpretive framework rather than as a measurement model.

2.3 The Competency-Based Curriculum in Kenya

Kenya's move to a competency-based curriculum was informed by concerns that the 8-4-4 system had become too examination-oriented and was not producing graduates with the practical skills and values needed in a changing economy (Akala, 2021). The Basic Education Curriculum Framework introduced a vision of nurturing every learner's potential, and set out seven core competencies, a set of national values and a range of pertinent and contemporary issues (PCIs), such as health, environmental awareness, citizenship and life skills, to be integrated across learning areas. Learners are expected to take part in activities that help them to discover, create and apply knowledge, while teachers act as facilitators rather than the sole source of information (Thukia, 2025).

Thukia (2025) analysed the reform using Basil Bernstein's ideas about curriculum and pedagogy and argued that the CBC represents a shift from a strongly framed, teacher-controlled model to a more weakly framed model in which learners have greater control over the pace and sequence of learning. The author notes, however, that such a shift places heavy demands on teachers, who must be confident enough to relinquish some control while still guiding learning purposefully. Where teachers lack this confidence or the necessary skills, the result can be a hybrid in which new terminology is adopted but old routines persist.

Assessment is a central pillar of the reform. The KNEC (2021) framework requires teachers to carry out school-based assessment using a variety of tools and to record learner progress against four levels of achievement: exceeding expectations, meeting expectations, approaching expectations and below expectations. The PWPER (2023) acknowledged that the implementation of the curriculum had been hampered by inadequate teacher preparation, an overloaded timetable and insufficient resources, and recommended that the number of learning areas be reduced and that teacher capacity building be intensified. These recommendations confirm that teacher competence is widely recognised as a decisive factor in the success of the reform. They also show that the curriculum is still evolving, which means teachers must continually update their knowledge.

2.4 Empirical Review

This section reviews empirical studies related to each of the four objectives of the study. Studies from other countries are reviewed first where relevant, followed by studies from Kenya, so that the reader can see how the issues play out at different levels.

2.4.1 Teachers' Knowledge and Understanding of the Competency-Based Curriculum

A teacher who does not understand what a curriculum is trying to achieve can hardly implement it well. International evidence consistently shows that teachers' understanding of competence-based reforms is often partial. In their scoping review of competence-based curriculum implementation in seven African countries, Wawire et al. (2026) found that many teachers equated the new curriculum with a set of new activities or a new lesson plan template, without grasping the deeper shift from content coverage to competence development. In Tanzania, Kasuga and Kalolo (2025) reported that teachers in Njombe Region expressed uncertainty about what exactly counted as a competence and how it should be assessed, which led them to fall back on traditional practices. Similar observations were made in Uganda, where Kidega et al. (2024) found that teachers in Gulu City struggled to interpret the new curriculum's generic skills and project-based learning requirements.

In Kenya, early studies set the tone. Mwang'ombe (2021) examined teachers' understanding and skills in relation to CBC and concluded that their understanding was often vague and uneven. In practice, such vagueness can mean that a teacher is able to name the core competencies without being sure how to develop them in a particular lesson. Isaboke et al. (2021), working in public pre-primary schools in Nairobi City County, found that 65.9% of teachers had not received any training on CBC. The authors linked this gap in training, together with inadequate facilities and large classes, to weak implementation. The study used a population of 450 pre-primary teachers, 225 ECD centre managers and 225 head teachers, and its findings are striking because Nairobi is among the best-resourced counties in the country.

Later studies have produced mixed results. Mutua and Waweru (2023), studying public primary schools in Machakos County, reported that yearly in-service training for teachers was considered adequate. However, they found that constraints such as a shortage of learning materials and a high pupil-teacher ratio limited the translation of knowledge into practice. Karua et al. (2025) examined teachers' awareness of the tenets of CBC in Kinangop Sub-County and found that while most teachers had heard of the core competencies and values, a majority (52.3%) were not competent in preparing

schemes of work in line with the curriculum. This suggests that knowledge in the sense of awareness may coexist with poor procedural knowledge.

Mwangi et al. (2026) offered another useful perspective. Although 95% of the teachers in their study had been trained, 65% held negative perceptions of the curriculum. Their findings suggest that training programmes may have focused on transmitting information rather than building deep understanding and ownership. This aligns with the view of Jentsch and König (2022) that teacher competence involves beliefs and motivation, and not just information. In junior secondary schools in Bomet County, Keter and Wabuke (2025) found that teachers' preparedness, including their understanding of the curriculum, was significantly associated with the level of implementation, and that gaps in preparation continued to limit the quality of implementation.

Few studies have assessed teachers' knowledge in the ASAL counties. Hussein (2023) reported that teachers in northern Kenya had limited exposure to CBC training and lacked a clear grasp of competency-based grading, while Ongubo (2025) found that over 60% of public primary teachers in Township Ward, Garissa, had not received comprehensive in-service training. Neither study, however, assessed teachers' knowledge in a systematic way using a knowledge scale or linked it statistically to implementation outcomes. The present study will address this gap in Habaswein.

2.4.2 Teachers' Pedagogical Competence in Implementing the CBC

Pedagogical competence refers to what teachers do in the classroom: how they plan, how they organise learning activities, how they engage learners and how they use resources. The CBC expects teachers to use learner-centred approaches such as inquiry, cooperative learning, role play, projects, field visits and problem solving, and to integrate values and PCIs into lessons. Wawire et al. (2026) observed that across African countries implementing competence-based curricula, lessons were still frequently dominated by teacher explanations, choral responses and note-taking, with limited opportunities for learners to collaborate, investigate or create. They attributed this to large classes, a shortage of materials and gaps in teacher preparation, but also to entrenched habits formed under earlier examination-oriented systems.

In Kenya, the classroom-based study by APHRC (2022) provides one of the most detailed pictures of pedagogical practice under CBC. The study evaluated the quality of CBC implementation by examining lessons in schools and sought to identify the features of effective lessons, such as clear learning outcomes, active learner engagement and deliberate integration of core competencies. Its evidence suggests that effective CBC lessons are possible within Kenyan public schools, but that their quality

varies considerably, with many lessons only partly reflecting the curriculum's expectations. Mohamed et al. (2022), studying geography teachers in Kirinyaga County, found that teachers' pedagogical preparedness was low, with many teachers having limited experience with learner-centred methods and inadequate exposure to CBC-oriented training.

Alwang'a and Were (2026) examined the status of teachers' pedagogical knowledge and its link with CBC implementation in public primary schools in Sabatia Sub-County, Vihiga County. Their study linked the state of teachers' pedagogical knowledge to the manner in which the curriculum was being implemented, underlining pedagogical knowledge as a key ingredient of effective CBC delivery. Karua et al. (2025) found that teachers in Kinangop had a low capacity to select appropriate teaching and learning approaches for CBC classes, while Mutai et al. (2024) reported, from a survey of 605 respondents in Moiben Sub-County, that teacher preparedness in terms of pedagogical skills had a significant influence on CBC implementation.

Assessment is an integral part of pedagogy under CBC. Macheso, Kati, and Wafula (2024) found that teachers in western Kenya used written tests and oral questions far more often than portfolios, projects and anecdotal records, even though the KNEC (2021) framework expects a wide range of tools. In a related study of Grade 6 science and technology teachers in Bungoma, Busia and Trans-Nzoia, Macheso, Wafula, and Kati (2024) found that teachers' self-efficacy was associated with the extent to which they implemented competency-based assessment. In other words, teachers who felt confident about assessment tended to implement competency-based assessment more fully. These findings are important because they show that pedagogical competence is shaped not only by skill but also by teachers' belief in their own ability, which is consistent with the Teacher Professional Competence Model.

Digital pedagogy is another dimension. Murithi and Yoo (2021) surveyed 351 teachers and found that although teachers viewed ICT as necessary for CBC, most had received only basic computer literacy training and struggled to integrate technology into lessons. In a place like Habaswein, where connectivity and electricity are limited, the challenge is likely to be even greater. It remains unclear how teachers in such settings interpret and implement the digital literacy competency, and whether they use alternative low-cost approaches.

2.4.3 Factors Affecting Teachers' Competence in Implementing the CBC

The literature identifies several groups of factors that shape teachers' competence. These can be grouped into training and professional development, instructional

resources and infrastructure, workload and staffing, supervision and support, attitudes and beliefs, and contextual conditions.

The first and most frequently cited factor is training. UNESCO and the International Task Force on Teachers for Education 2030 (2024) argue that professional development must be continuous, practice-based and responsive to teachers' needs if it is to improve teaching. In Kenya, however, the initial CBC training was widely described as brief and rushed. Okeyo and Mokuia (2023) concluded that training should not be hurriedly done as a public relations exercise, after finding that most teachers in Nyamira South were ill-prepared. Ongubo (2025) reported that in Garissa, the reach of in-service training was limited, and that teachers who missed the initial cascade sessions had few opportunities to catch up. Mogere (2022), in a study of public primary schools in Nairobi County, likewise identified teacher training as an important determinant of how well the CBC was being implemented.

The second group of factors relates to instructional resources and infrastructure. Cheruiyot (2024) identified a shortage of learning materials, inadequate classrooms and limited technological infrastructure as major challenges in junior schools, alongside resistance to change among teachers, parents and learners. Hussein (2023) found that poor infrastructure and a lack of learning resources undermined proper assessment in northern Kenya. Mutua and Waweru (2023) similarly reported that inadequate teaching and learning materials hindered implementation in Machakos. When resources are scarce, even competent teachers may be unable to use the methods they know to be effective.

Third, workload and staffing affect how much time and energy teachers can devote to planning, assessment and reflection. The CBC requires detailed lesson planning and continuous assessment records for every learner, which can be overwhelming in large classes. Isaboke et al. (2021) listed large class sizes and understaffing among the major barriers in Nairobi, while Mutua and Waweru (2023) highlighted the high pupil-teacher ratio. In ASAL counties, the shortage of teachers is more acute. The County Government of Wajir (2023) acknowledges that teacher shortages, compounded by the transfer of non-local teachers due to insecurity, remain a serious challenge in the county's schools. In such conditions, schools often depend on Board of Management teachers who may not have received any formal CBC training.

Fourth, supervision and support from head teachers and Curriculum Support Officers influence the development of teacher competence. Awiti (2023), in a study of primary schools in Nairobi City, examined a range of school-level factors affecting CBC implementation and pointed to the importance of the support that teachers receive

within their schools. In principle, head teachers who regularly observe lessons, review schemes of work and organise peer learning sessions create the conditions in which teachers can grow in confidence. Keter and Wabuke (2025) likewise emphasised the role of continuous support in strengthening teacher preparedness.

Fifth, attitudes and beliefs matter. Mwangi et al. (2026) found that a majority of teachers held negative perceptions of CBC despite being trained, while Cheruiyot (2024) identified resistance to change among teachers as a barrier. The Teacher Professional Competence Model predicts that such beliefs will affect how teachers perceive classroom situations and how they respond to them. Teachers who doubt the value of learner-centred methods may revert to lecturing, especially when under pressure to cover the syllabus.

Finally, contextual conditions in marginalised areas deserve attention. Barasa et al. (2025) showed in Uganda that schools serving low-income communities struggled more with competence-based reform than schools serving wealthier families, and warned that without deliberate support the reform could entrench inequality. Kasuga and Kalolo (2025) made a similar point about rural schools in Tanzania. In northern Kenya, drought, pastoralist mobility, long distances and insecurity disrupt both learner attendance and teachers' access to training (County Government of Wajir, 2023; Hussein, 2023). These conditions may lower teacher morale and increase turnover, weakening the continuity required for competence to develop. Very few studies have explored how these contextual factors interact with teacher competence in the Kenyan CBC context, and this study will seek to do so.

2.4.4 Teachers' Competence and Effective Implementation of the CBC

The final objective focuses on the relationship between teachers' competence and the effectiveness of CBC implementation. Several Kenyan studies have explored related relationships, though often using preparedness rather than competence as the independent variable. Mutai et al. (2024) found a significant positive relationship between teacher preparedness and CBC implementation in Moiben Sub-County. Keter and Wabuke (2025) reported that teacher preparedness significantly influenced the implementation of the curriculum in public junior secondary schools in Bomet County. Mutua and Waweru (2023) examined teachers' competence in terms of values, attitudes, knowledge and skills alongside the extent of CBC implementation in Machakos County, and concluded that implementation depended both on teachers and on the conditions under which they worked, such as the supply of materials and the pupil-teacher ratio.

Alwang'a and Were (2026) went further by focusing on pedagogical knowledge as a specific dimension of competence. Their work suggests that the status of teachers' pedagogical knowledge has a direct bearing on the implementation of the CBC in Sabatia, which strengthens the case for targeted professional development in learner-centred strategies. Macheso, Wafula, and Kati (2024) found that teacher efficacy was linked to the implementation of competency-based assessment. The APHRC (2022) evaluation likewise suggests that the quality of CBC lessons depends heavily on what teachers bring to the classroom, including their planning and their understanding of the intended competencies.

Outside Kenya, Wawire et al. (2026) concluded that competence-based reforms in Africa were more likely to succeed where teachers had sustained professional support, adequate resources and a clear understanding of the reform's goals. Kidega et al. (2024) similarly recommended sustained training and school-level support for teachers in Uganda, noting that the challenges teachers faced were closely tied to gaps in their preparation for the new curriculum. Jentsch and König (2022), drawing on international studies, provide the theoretical rationale for these findings: teacher competence influences the quality of instruction, which in turn influences learner outcomes.

Despite this growing evidence, three weaknesses remain. First, most of the studies measured competence or preparedness through self-report questionnaires alone, which may inflate estimates of competence. Second, few studies distinguished between different dimensions of competence and tested their separate contributions to implementation. Third, almost none of these studies were conducted in the ASAL counties of North Eastern Kenya. The present study will address these weaknesses by combining questionnaires with classroom observation and interviews, by testing the separate and combined contributions of knowledge and pedagogical competence, and by focusing on Habaswein Sub-County.

2.5 Critique of Existing Literature

The literature reviewed shows broad agreement that teacher competence is central to the success of the CBC, and that many Kenyan teachers have not yet acquired the full range of competences required. Nonetheless, several weaknesses can be identified. To begin with, there is a strong geographical bias. Studies have been concentrated in Nairobi, central Kenya, the Rift Valley and western Kenya (Isaboke et al., 2021; Karua et al., 2025; Mohamed et al., 2022; Mutai et al., 2024). The few studies conducted in northern Kenya, such as Hussein (2023) and Ongubo (2025), were largely descriptive and did not test relationships between competence and implementation.

In addition, there is a conceptual inconsistency in the literature. Terms such as preparedness, readiness, competence and capacity are often used interchangeably, and many studies measured preparedness mainly by asking whether teachers had attended training. Mwangi et al. (2026) demonstrate that training attendance is a poor proxy for competence, since trained teachers may still lack understanding or hold negative attitudes. The present study adopts a clearer conception of competence based on the Teacher Professional Competence Model (Jentsch & König, 2022), distinguishing knowledge from pedagogical competence.

There is also a methodological weakness. Many studies relied solely on questionnaires, which are prone to social desirability bias. Only a few, such as APHRC (2022) and Okeyo and Mokua (2023), used observation schedules or document analysis to verify teachers' reported practice. Furthermore, some studies used small samples or purposive sampling that limit generalisation. For instance, Mohamed et al. (2022) drew their conclusions from only 40 respondents. The present study will use stratified random sampling, an adequate sample size and triangulation of data sources to strengthen the credibility of its findings.

Finally, the literature has tended to treat contextual factors as background rather than as variables that interact with teacher competence. Barasa et al. (2025) and Kasuga and Kalolo (2025) show that context shapes how competence-based reforms unfold, yet Kenyan studies have rarely explored how factors such as drought, insecurity and pastoralist mobility affect teachers' professional growth. The present study will examine these factors explicitly.

2.6 Conceptual Framework

The conceptual framework presents the hypothesised relationship between the independent, intervening and dependent variables of the study, as shown in Figure 2.1.

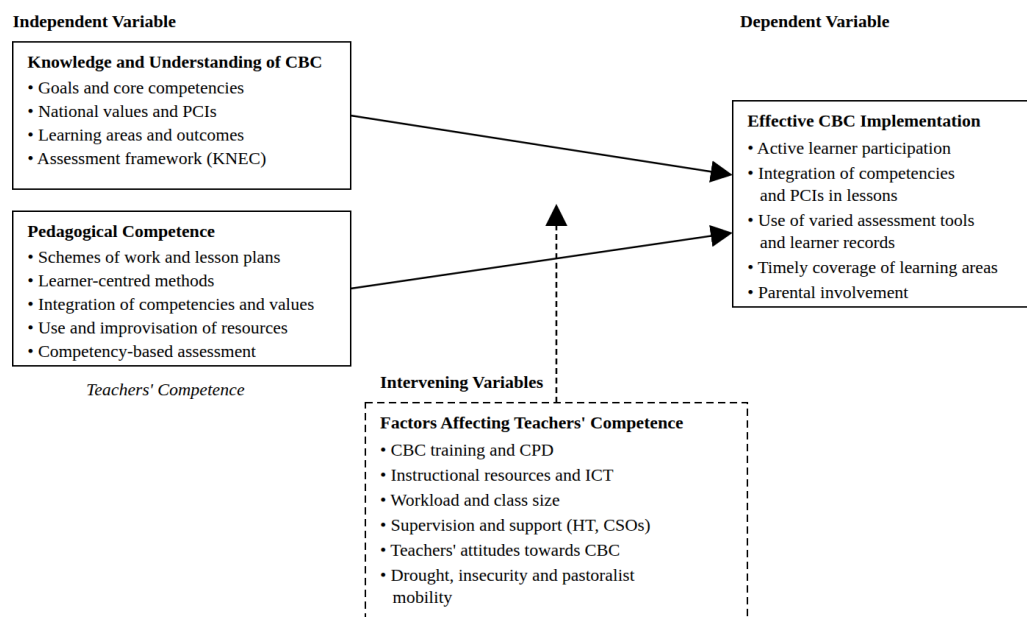


Figure 2.1: Conceptual Framework

Source: Researcher (2026)

The independent variable is teachers' competence, which is broken down into two dimensions. The first is knowledge and understanding of the CBC, reflected in teachers' grasp of the curriculum's goals, core competencies, values, PCIs, learning areas and assessment requirements. The second is pedagogical competence, reflected in teachers' ability to prepare schemes of work and lesson plans in the required format, use learner-centred methods, integrate core competencies and values, use and improvise resources, and carry out competency-based assessment.

The dependent variable is the effective implementation of the CBC. It will be measured by indicators that reflect the intentions of the curriculum: active learner participation, the integration of core competencies and PCIs in lessons, the use of varied assessment tools and proper learner records, coverage of learning areas as scheduled, and the involvement of parents in learning activities.

The factors affecting teachers' competence are treated as intervening variables. They include the adequacy of CBC training and continuous professional development, the availability of instructional resources and ICT infrastructure, teacher workload and class size, supervision and support from head teachers and Curriculum Support Officers, teachers' attitudes towards the CBC, and contextual conditions such as drought, insecurity and pastoralist mobility. These factors may strengthen or weaken the relationship between competence and implementation. For example, a knowledgeable teacher may still be unable to use group work effectively in a class of 90

learners without desks. Government policy, including the reforms recommended by the PWPER (2023), is recognised as a broader influence but will not be measured directly.

2.7 Summary of Literature and Research Gaps

The literature shows that the CBC places high demands on teachers, requiring a solid understanding of the curriculum, learner-centred pedagogy and competency-based assessment. The three theories reviewed help to explain these demands. CBAM shows that teachers adopt innovations gradually and need support tailored to their concerns; the Teacher Professional Competence Model shows that competence combines knowledge, beliefs and situation-specific skills; and TPACK highlights the integrated knowledge needed to teach content effectively with available tools in a specific context. Empirical studies in Kenya and elsewhere in Africa confirm that gaps in teacher competence are widespread and are shaped by training, resources, workload, supervision, attitudes and context. Table 2.1 summarises the key studies and the gaps they leave.

Table 2.1: Summary of Research Gaps

Author(s) and Year	Focus of the Study	Key Findings	Research Gap and How the Study Addresses It
Isaboke et al. (2021)	Teacher preparedness and CBC implementation in pre-primary schools, Nairobi	65.9% of teachers had not received CBC training; large classes and inadequate resources hindered implementation	Urban, pre-primary focus; current study targets primary schools in a marginalised ASAL setting
Okeyo and Mokuia (2023)	Preparedness of teachers for CBC in secondary schools, Nyamira South	Schools were ill-prepared; most teachers were untrained; CBAM used	Secondary school focus; current study examines primary schools and measures competence directly
Mutua and Waweru (2023)	Teachers' competence and CBC implementation, Machakos County	Training considered adequate, but materials and high pupil-teacher ratio constrained implementation	Relatively well-served county; current study examines an ASAL context with different constraints
Hussein (2023)	Challenges of CBC implementation in northern Kenya	Poor infrastructure, few resources and limited training undermined implementation	Descriptive study; current study tests the statistical relationship between competence and implementation

Author(s) and Year	Focus of the Study	Key Findings	Research Gap and How the Study Addresses It
Macheso, Kati, and Wafula (2024)	Use of competency-based assessment tools in western Kenya	Teachers relied on a narrow range of assessment tools	Focused on assessment only; current study includes assessment as part of overall pedagogical competence
Karua et al. (2025)	Teacher preparedness for CBC, Kinangop Sub-County	52.3% of teachers were not competent in preparing schemes of work	Lower primary focus in a highland county; current study covers Grades 1 to 6 in Wajir
Ongubo (2025)	Teacher preparedness for CBC adoption, Garissa County	Over 60% of teachers lacked comprehensive in-service training	Limited to one ward; did not link competence to implementation outcomes
Mwangi et al. (2026)	Teacher training and perceptions prior to CBC implementation	95% trained but 65% held negative perceptions	Did not assess classroom practice; current study uses observation to verify competence
Wawire et al. (2026)	Scoping review of CBC pedagogy and assessment across Africa	Classroom practice remains largely teacher-centred	Regional synthesis; current study provides primary data from a specific marginalised context

Source: Researcher (2026)

In summary, the reviewed literature reveals a contextual gap, since no known study has examined teacher competence and CBC implementation in Habaswein Sub-County or in Wajir County more generally. It also reveals a conceptual gap, because many studies have treated training attendance as a proxy for competence, and a methodological gap, since few studies have triangulated self-reported data with classroom observation or tested the separate contributions of different dimensions of competence. The present study is designed to address these gaps.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes how the study will be carried out. It presents the research design, the location of the study, the target population, the sampling techniques and sample size, the research instruments, the pilot study, the validity and reliability of the instruments, the data collection procedures, the data analysis techniques, the logistical arrangements and the ethical considerations that will guide the study.

3.2 Research Design

The study will adopt a convergent parallel mixed methods design, combining a descriptive survey with a correlational component. In a convergent design, quantitative and qualitative data are collected during the same phase, analysed separately and then merged during interpretation so that each type of data can confirm, extend or challenge the other (Creswell & Creswell, 2023). The descriptive survey element will allow the researcher to describe the current level of teachers' knowledge, pedagogical competence and the factors affecting their competence, as they exist, without manipulating any variable. The correlational element will allow the researcher to test the strength and direction of the relationship between teachers' competence and effective CBC implementation, as stated in the hypotheses.

This design is appropriate for three reasons. First, the study seeks both to describe a situation and to test relationships, which requires quantitative data from a relatively large number of teachers. Second, the concept of competence is complex, and self-reported questionnaire data alone may not capture it accurately; interviews with head teachers and Curriculum Support Officers, together with classroom observations, will provide richer and more credible evidence. Third, the design has been used successfully in similar studies of CBC implementation in Kenya. Okeyo and Mokua (2023), for example, used a convergent mixed methods design to study teacher preparedness in Nyamira South, and Mutua and Waweru (2023) used a descriptive survey to study teacher competence in Machakos. The design does, however, require careful planning to ensure that the two strands of data are properly integrated, and the researcher will address this through a joint display of findings during analysis.

3.3 Location of the Study

The study will be conducted in Habaswein Sub-County, one of the sub-counties of Wajir County in the North Eastern region of Kenya. The sub-county lies in the southern

part of the county, along the main road linking Garissa and Wajir towns, and it borders Garissa County to the south and west. The area is largely arid, with low and erratic rainfall, high temperatures and recurrent droughts. The majority of residents are pastoralists who keep camels, cattle, goats and sheep, and many households move seasonally in search of water and pasture. Habaswein town serves as the administrative and commercial centre of the sub-county, while the rest of the population lives in smaller settlements spread over a vast area (County Government of Wajir, 2023).

Education indicators in Wajir County are among the lowest in Kenya. The County Integrated Development Plan identifies low enrolment and transition rates, a shortage of teachers, inadequate classrooms and sanitation facilities, and the negative effects of drought and insecurity on schooling as key challenges (County Government of Wajir, 2023). Several public primary schools in the sub-county operate low-cost boarding facilities to keep children from mobile households in school. The choice of Habaswein is based on three considerations: it is representative of the marginalised ASAL context in which CBC implementation is thought to be most difficult; no previous study has examined teacher competence and CBC implementation there; and the researcher is familiar with the area, which will ease access and communication with respondents.

3.4 Target Population

The target population will comprise all public primary schools in Habaswein Sub-County, together with their head teachers and teachers, and the Curriculum Support Officers (CSOs) who serve the sub-county. The unit of analysis will be the school, while the units of observation will be the teachers, head teachers and CSOs. Teachers are included because they are the direct implementers of the CBC. Head teachers are included because they are responsible for instructional leadership, supervision and the allocation of resources at school level. CSOs are included because they provide professional support to teachers and monitor curriculum implementation across schools.

According to preliminary information gathered by the researcher, the sub-county has approximately 42 public primary schools with an estimated 336 teachers, including both TSC and Board of Management teachers, and 3 CSOs. These figures are provisional and will be confirmed with the Sub-County Director of Education and the TSC Sub-County Director before data collection begins. The distribution of the target population is shown in Table 3.1.

Table 3.1: Target Population

Category of School (Stratum)	Number of Schools	Number of Head Teachers	Number of Teachers
Schools in Habaswein town and its environs	12	12	120
Schools in rural settlements (including low-cost boarding schools)	30	30	216
Total	42	42	336

Source: Provisional figures, to be confirmed with the Habaswein Sub-County Education Office (2026)

In addition to the population in Table 3.1, the 3 Curriculum Support Officers serving Habaswein Sub-County will form part of the target population.

3.5 Sampling Techniques and Sample Size

3.5.1 Sampling Techniques

The study will use a combination of stratified random sampling, simple random sampling, proportionate allocation and census methods. Schools will first be divided into two strata according to their location: schools in Habaswein town and its environs, and schools in rural settlements. Stratification is necessary because schools in and around the town are likely to have better access to transport, electricity, training venues and supervision than rural schools, and these differences may affect teacher competence. Within each stratum, schools will be selected through simple random sampling using a table of random numbers or a computer-generated list, which gives every school an equal chance of selection and reduces selection bias (Creswell & Creswell, 2023).

Half of the schools in each stratum, that is 6 schools from the town stratum and 15 from the rural stratum, will be selected, giving a total of 21 schools. Selecting 50% of schools is considered adequate for a relatively small population of schools and allows the researcher to cover both strata meaningfully within the available time and budget. The head teachers of the 21 sampled schools will be included automatically in the sample, since each school has only one head teacher. All 3 CSOs will be included through census because their number is small and their perspective is important for understanding the support available to teachers.

3.5.2 Sample Size

The sample size of teachers will be determined using the following finite population formula, which is commonly applied in educational surveys when the population is known:

$$n = N / (1 + N(e)^2)$$

Where n is the required sample size, N is the target population of teachers (336) and e is the level of precision, set at 0.05 for a 95% confidence level. Substituting these values gives:

$$n = 336 / (1 + 336(0.05)^2) = 336 / 1.84 = 182.6 \approx 183 \text{ teachers}$$

The sample of 183 teachers will be distributed proportionately between the two strata using the formula: stratum sample = (stratum population / total population) $\times$ 183. Within each sampled school, teachers will be listed from the staff register and selected through simple random sampling until the required number for that stratum is reached. Where a sampled school has fewer teachers than required, all its teachers will be included and the balance will be drawn from other sampled schools in the same stratum. The resulting sample distribution is shown in Table 3.2.

Table 3.2: Sample Size Distribution

Category of Respondents	Target Population	Sample Size	Sampling Technique
Teachers: Habaswein town and environs	120	65	Stratified, proportionate and simple random
Teachers: Rural settlements	216	118	Stratified, proportionate and simple random
Head teachers	42	21	Head teachers of the sampled schools
Curriculum Support Officers	3	3	Census
Total	381	207	

Source: Researcher (2026)

In addition, 21 CBC lessons, one in each sampled school, will be observed using a classroom observation checklist. The teacher to be observed in each school will be randomly selected from among the sampled teachers who teach Grades 4 to 6, where most of the CBC upper primary learning areas and school-based assessments are concentrated. Participation in the observation will be voluntary.

3.6 Research Instruments

Three instruments will be used to collect data: a questionnaire for teachers, an interview guide for head teachers and CSOs, and a classroom observation checklist. Using multiple instruments will allow the researcher to triangulate the data and to strengthen the credibility of the findings.

3.6.1 Questionnaire for Teachers

The teachers' questionnaire (Appendix III) will be the main instrument for collecting quantitative data. It will be divided into six sections. Section A will collect demographic information, including gender, age, academic and professional qualifications, teaching experience, employer (TSC or Board of Management), grades taught and CBC training attended. Section B will measure teachers' knowledge and understanding of the CBC. Section C will measure pedagogical competence, including planning, learner-centred methods, integration of competencies and assessment. Section D will focus on the factors affecting teachers' competence. Section E will measure the effectiveness of CBC implementation. Section F will contain open-ended questions that give teachers the opportunity to express their views in their own words.

Sections B to E will use five-point Likert scale items ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). Section B will also include a short set of knowledge items that require teachers to identify correct statements about the CBC, so that the measurement of knowledge does not rely entirely on self-rating. Questionnaires are appropriate for this study because they allow data to be collected from a large number of respondents within a short time, they guarantee anonymity, and the data can be easily coded and analysed statistically (Creswell & Creswell, 2023). The questionnaire will be administered in English, which is the medium of instruction and the official working language of teachers.

3.6.2 Interview Guide for Head Teachers and Curriculum Support Officers

Semi-structured interviews will be conducted with the 21 head teachers and 3 CSOs using the interview guide in Appendix IV. The guide will explore their views on teachers' knowledge and pedagogical practice, the support available to teachers, the challenges faced in implementing the CBC and their suggestions for improvement. Semi-structured interviews allow the researcher to probe for detail and clarification while still keeping the conversation focused on the study objectives. Each interview is expected to last between 30 and 45 minutes. With the permission of the interviewee, interviews will be audio-recorded and later transcribed; where permission is not given, detailed notes will be taken.

3.6.3 Classroom Observation Checklist

A structured classroom observation checklist (Appendix V) will be used to record what actually happens during CBC lessons. It will capture evidence of lesson planning, the use of learner-centred methods, the integration of core competencies, values and PCIs, the use of teaching and learning resources, learner participation and assessment practices. Each item will be rated on a scale from 1 (Not observed) to 4 (Fully observed), and the observer will also write brief field notes. Observation will help to verify the practices that teachers report in the questionnaire and will provide direct evidence of implementation. To reduce the effect of the observer's presence, the researcher will spend a few minutes in the class before the lesson begins and will sit at the back of the room without interfering.

3.6.4 Operationalisation of Study Variables

Table 3.3 shows how each variable will be measured and analysed.

Table 3.3: Operationalisation of Study Variables

Variable	Indicators	Measurement Scale	Instrument	Analysis Technique
Knowledge and understanding of CBC (Independent)	Understanding of goals, core competencies, values, PCIs, learning areas and assessment framework	Ordinal (Likert); knowledge test score	Questionnaire ; interview guide	Frequencies, percentages, means, standard deviation; Pearson correlation; regression
Pedagogical competence (Independent)	Schemes of work and lesson plans; learner-centred methods; integration of competencies; use of resources; competency-based assessment	Ordinal (Likert); observation rating	Questionnaire ; observation checklist	Means, standard deviation; Pearson correlation; regression
Factors affecting competence (Intervening)	Training and CPD; resources and ICT; workload and class size; supervision; attitudes; drought, insecurity and mobility	Ordinal (Likert); open-ended responses	Questionnaire ; interview guide	Frequencies, percentages, means; thematic analysis
Effective CBC implementation (Dependent)	Learner participation; integration of competencies and PCIs; use of varied assessment tools and records; coverage of learning areas; parental involvement	Ordinal (Likert); observation rating	Questionnaire ; observation checklist; interview guide	Means, standard deviation; regression; thematic analysis

Source: Researcher (2026)

3.7 Pilot Study

Before the main study, the instruments will be pre-tested in a pilot study in neighbouring Wajir South Sub-County, which shares similar climatic, social and educational characteristics with Habaswein but will not form part of the main study. Approximately 10% of the sample will be used in the pilot, which translates into 18 teachers, 2 head teachers and 2 classroom observations. The pilot will help the researcher to identify ambiguous, repetitive or sensitive questions, estimate the time required to complete each instrument, check the clarity of instructions and test the data entry and coding procedures. The feedback from the pilot will be used to revise the instruments before the main data collection.

3.7.1 Validity of the Research Instruments

Validity refers to the extent to which an instrument measures what it is intended to measure. The study will focus on content validity and construct validity. To establish content validity, the instruments will be reviewed by the researcher's supervisors and two experts in curriculum studies at Mount Kenya University. Each expert will rate the relevance of every item to the study objectives on a four-point scale, and a Content Validity Index (CVI) will be calculated by dividing the number of items rated relevant or highly relevant by the total number of items. An index of 0.78 or above will be considered acceptable, and items falling below this threshold will be revised or removed. Construct validity will be enhanced by basing the questionnaire items on the indicators derived from the theories and literature reviewed in Chapter Two, and by checking during the pilot that items intended to measure the same construct are correlated with one another.

3.7.2 Reliability of the Research Instruments

Reliability refers to the consistency with which an instrument produces the same results under similar conditions. The internal consistency of the Likert-scale sections of the teachers' questionnaire will be tested using Cronbach's alpha coefficient, computed from the pilot data. A coefficient of 0.70 or higher will be considered acceptable, as recommended in social science research (Field, 2024). Where a section records a lower coefficient, the item-total statistics will be examined and weak items revised or removed. To improve the reliability of the observation checklist, the researcher and one trained research assistant will observe the same pilot lessons independently, and their ratings will be compared. Differences will be discussed and the descriptors on the checklist clarified until a high level of agreement is achieved.

3.7.3 Trustworthiness of Qualitative Data

For the qualitative data, the study will observe the criteria of credibility, dependability, confirmability and transferability. Credibility will be enhanced through triangulation of data from teachers, head teachers, CSOs and classroom observations, and through member checking, in which a sample of interviewees will be given summaries of their interviews to confirm that their views have been accurately captured. Dependability will be supported by keeping an audit trail of all decisions made during data collection and analysis. Confirmability will be enhanced by reflexive notes in which the researcher records personal assumptions that might influence interpretation. Transferability will be supported by providing a detailed description of the study context so that readers can judge whether the findings apply to similar settings (Braun & Clarke, 2022).

3.8 Data Collection Procedures

The researcher will first obtain an introduction letter from the School of Education at Mount Kenya University and ethical approval from the University's Ethics Review Committee. A research permit will then be sought from the National Commission for Science, Technology and Innovation (NACOSTI). Upon receiving the permit, the researcher will report to the Wajir County Commissioner and the County Director of Education to seek authority to conduct research in the county, and then to the Sub-County Director of Education and the TSC Sub-County Director in Habaswein. Head teachers of the sampled schools will be contacted in advance to explain the purpose of the study and to agree on convenient dates for the visits.

Two research assistants, who are qualified teachers familiar with the area and fluent in both English and Somali, will be recruited and trained for two days on the study objectives, the instruments, ethical procedures and security precautions. During school visits, the researcher and assistants will introduce themselves to the head teacher, explain the study and seek informed consent from participants. Questionnaires will be administered to the sampled teachers and collected on the same day wherever possible, to maximise the response rate. Where teachers request more time, the questionnaires will be collected within three days. Interviews with head teachers will be held in a private room at the school, while interviews with CSOs will be held at their offices. Classroom observations will be conducted at times agreed with the teachers. Data collection is expected to take about six weeks and will be scheduled during school terms, avoiding examination periods and, as far as possible, the peak of the dry season when some families move away.

3.9 Data Analysis and Presentation

3.9.1 Quantitative Data Analysis

After data collection, the questionnaires and observation checklists will be checked for completeness and consistency, coded and entered into the Statistical Package for the Social Sciences (SPSS), version 29. Data cleaning will involve checking for missing values, outliers and entry errors. Descriptive statistics, including frequencies, percentages, means and standard deviations, will be used to describe the demographic characteristics of respondents and to answer the first three research questions. Mean scores on the Likert items will be interpreted as follows: 1.00 to 1.80 very low, 1.81 to 2.60 low, 2.61 to 3.40 moderate, 3.41 to 4.20 high and 4.21 to 5.00 very high.

To test the hypotheses under the fourth objective, inferential statistics will be used. Pearson's product-moment correlation coefficient will be used to determine the strength and direction of the relationship between each dimension of teachers' competence and effective CBC implementation (H01 and H02). Multiple linear regression analysis will then be used to establish the combined and separate influence of the dimensions of competence on effective implementation (H03). The regression model will take the following form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Where Y represents effective CBC implementation, β_0 is the constant, β_1 and β_2 are the regression coefficients, X_1 is teachers' knowledge and understanding of the CBC, X_2 is teachers' pedagogical competence and ε is the error term. Composite scores for each variable will be computed as the mean of the relevant items. Before running the regression, the researcher will check the assumptions of normality, linearity, homoscedasticity, independence of errors and absence of multicollinearity, using histograms, scatter plots, the Durbin-Watson statistic and variance inflation factors (Field, 2024). The coefficient of determination (R^2) will show the proportion of variation in implementation explained by teachers' competence, while the F-test will indicate the overall significance of the model. Hypotheses will be tested at the 0.05 level of significance. The results will be presented in tables, bar charts and pie charts, accompanied by interpretation and discussion.

3.9.2 Qualitative Data Analysis

The qualitative data from interviews, open-ended questionnaire items and observation field notes will be analysed using reflexive thematic analysis, following the six phases described by Braun and Clarke (2022): familiarisation with the data, generating initial codes, constructing themes, reviewing themes, defining and naming themes, and

writing up. Interview recordings will first be transcribed verbatim, and any responses given in Somali will be translated into English and checked by a second translator for accuracy. The transcripts will then be read several times, coded and organised into themes that relate to the study objectives. NVivo software will be used to organise the codes and themes. The findings will be presented in narrative form, supported by direct quotations from participants, identified only by codes such as HT1 for the first head teacher and CSO2 for the second Curriculum Support Officer.

3.9.3 Integration of Quantitative and Qualitative Findings

In line with the convergent design, the quantitative and qualitative results will be merged during interpretation. The researcher will prepare a joint display table showing, for each objective, the main statistical results alongside the corresponding qualitative themes and observation evidence. Areas of agreement will strengthen confidence in the findings, while areas of disagreement will be explored and explained, for instance where teachers report a higher level of competence than is observed in lessons (Creswell & Creswell, 2023).

3.10 Logistical Considerations

The study will require careful logistical planning because of the vast distances and security situation in the sub-county. The researcher will hire a reliable vehicle for fieldwork and will plan the visits so that schools in the same area are covered on the same day. Movement will be restricted to daylight hours, and the researcher will consult the local administration and security officers before visiting remote schools. Research assistants will be drawn from the local community to ease communication and build trust. A detailed work plan and budget for the study are provided in Appendices VI and VII.

3.11 Ethical Considerations

The study will observe the ethical standards required for research involving human participants. Ethical approval will be sought from the Mount Kenya University Ethics Review Committee, and a research permit will be obtained from NACOSTI before data collection begins. Permission will also be sought from the county and sub-county education authorities and from the head teachers of the sampled schools.

Participation in the study will be voluntary. Each participant will receive an information sheet and a consent form (Appendix II) explaining the purpose of the study, what participation involves, the expected duration and the right to withdraw at any stage without penalty. Written informed consent will be obtained before any questionnaire is

administered, interview conducted or lesson observed. Teachers will be assured that the classroom observation is not an appraisal and that the results will not be shared with their employer or used for any disciplinary purpose.

Confidentiality and anonymity will be protected. Respondents will not be required to write their names on the questionnaires, and schools and participants will be identified by codes in all reports. Audio recordings, transcripts and completed instruments will be stored securely, with electronic files protected by passwords, in line with the Data Protection Act of 2019. The data will be used only for academic purposes and will be destroyed after the period required by the university. The researcher will also be sensitive to the cultural and religious norms of the community, including dress codes, prayer times and gender norms, and will ensure that interviews are conducted at times and places that are convenient and acceptable to participants. Finally, the researcher will acknowledge all sources used in the study through proper citation and referencing in APA style, and the final report will be checked for plagiarism in accordance with university policy.

REFERENCES

- African Population and Health Research Center. (2022). *Evaluating the quality of implementation of competency-based curriculum in Kenya: Evidence of effective lessons*. APHRC. <https://aphrc.org/idea/evaluating-the-quality-of-implementation-competency-based-curriculum-in-kenya-evidence-of-effective-lessons/>
- Akala, B. M. (2021). Revisiting education reform in Kenya: A case of Competency Based Curriculum (CBC). *Social Sciences & Humanities Open*, 3(1), Article 100107. <https://doi.org/10.1016/j.ssaho.2021.100107>
- Alwang'a, A. B., & Were, D. (2026). Status of teachers' pedagogical knowledge and the implementation of competency-based curriculum in public primary schools in Sabatia Sub-County, Kenya. *East African Journal of Education Studies*, 9(1), 367–381. <https://doi.org/10.37284/eajes.9.1.4414>
- Awiti, C. M. (2023). *Factors affecting implementation of competency-based curriculum of education in primary schools in Nairobi City* [Master's research project, University of Nairobi]. University of Nairobi Digital Repository. <https://erepository.uonbi.ac.ke/handle/11295/167233>
- Barasa, M. C., Ssentamu, P. N., Chang'ach, J. K., & Kurgat, S. J. (2025). Implementing the Uganda lower secondary competence-based curriculum: The equity question. *Quality Education for All*, 2(1), 138–157. <https://doi.org/10.1108/QEA-03-2024-0026>
- Braun, V., & Clarke, V. (2022). *Thematic analysis: A practical guide*. SAGE.
- Cheruiyot, B. (2024). Challenges faced in the implementation of competency-based curriculum (CBC) in junior schools in Kenya. *East African Journal of Education Studies*, 7(3), 260–266. <https://doi.org/10.37284/eajes.7.3.2098>
- County Government of Wajir. (2023). *Wajir County integrated development plan 2023–2027*. <https://maarifa.cog.go.ke/sites/default/files/2024-06/WAJIR%20CIDP%202023%20-%202027.pdf>
- Creswell, J. W., & Creswell, J. D. (2023). *Research design: Qualitative, quantitative, and mixed methods approaches* (6th ed.). SAGE.
- Field, A. (2024). *Discovering statistics using IBM SPSS Statistics* (6th ed.). SAGE.
- Hussein, A. M. (2023). The challenge face in the implementation of competency-based curriculum in northern Kenya. *African Journal of Education and Practice*, 9(3), 59–67. <https://iprjb.org/journals/AJEP/article/view/2163>

- Isaboke, H., Mweru, M., & Wambiri, G. (2021). Teacher preparedness and implementation of the competency based curriculum in public pre-primary schools in Nairobi City County, Kenya. *International Journal of Current Aspects*, 5(3), 32–53. <https://doi.org/10.35942/ijcab.v5i3.186>
- Jentsch, A., & König, J. (2022). Teacher competence and professional development. In T. Nilsen, A. Stancel-Piątak, & J.-E. Gustafsson (Eds.), *International handbook of comparative large-scale studies in education: Perspectives, methods and findings*. Springer. https://doi.org/10.1007/978-3-030-88178-8_38
- Karua, G. W., Manyasi, B., & Kara, A. (2025). Teacher preparedness to implement competency based curriculum in public primary schools in Kinangop Sub County, Kenya. *International Journal of Scientific Research and Management*, 13(2), 22–41. <https://www.ijstrm.net/index.php/ijstrm/article/view/6027>
- Kasuga, W., & Kalolo, J. (2025). Competency-based curriculum in Tanzania: Charting dilemmas for successful implementation. *Cogent Education*, 12(1), Article 2563711. <https://doi.org/10.1080/2331186X.2025.2563711>
- Kenya National Examinations Council. (2021). *Competency based assessment framework*. KNEC. <https://www.knec.ac.ke/wp-content/uploads/2021/06/CBA-Age-Based-Regular-Framework.pdf>
- Keter, J. K., & Wabuke, J. M. (2025). Teacher preparedness and implementation of competency-based curriculum in public junior secondary schools in Bomet County, Kenya. *International Journal of Social Science and Economics Invention*, 11(7), 116–123. <https://doi.org/10.23958/ijsssei/vol11-i7/404>
- Kidega, C., Song, Z., Ugochinyere, I. C., James, O. A., & Ndikubwimana, F. (2024). Confronting challenges facing teachers in implementing competency-based curriculum in Uganda: A case of secondary schools in Gulu City. *East African Journal of Education Studies*, 7(2), 112–129. <https://doi.org/10.37284/eajes.7.2.1891>
- Macheso, I. S., Kati, R. O., & Wafula, R. W. (2024). Implementation of competency based assessment in Kenya: Extent of utilization of competency based assessment tools and types. *International Journal of Research and Innovation in Social Science*, 8(6), 3124–3134. <https://doi.org/10.47772/IJRISS.2024.806236>

- Macheso, I. S., Wafula, R. W., & Kati, R. O. (2024). Teacher efficacy and implementation of competency based assessment in Western Kenya: A case of Grade 6 science and technology teachers in Bungoma, Busia and Trans-Nzoia Counties. *International Journal of Research and Innovation in Applied Science*, 9(8). <https://doi.org/10.51584/IJRIAS.2024.908011>
- Mishra, P., Warr, M., & Islam, R. (2023). TPACK in the age of ChatGPT and generative AI. *Journal of Digital Learning in Teacher Education*, 39(4), 235–251. <https://doi.org/10.1080/21532974.2023.2247480>
- Mogere, V. B. (2022). *Determinants of competency-based curriculum implementation in public primary schools in Nairobi County* [Master's thesis, Kenyatta University]. Kenyatta University Institutional Repository. <https://ir-library.ku.ac.ke/items/145b97f8-a06b-4cad-ac5d-16892512ddfb>
- Mohamed, M. M., Ondigi, S. R., & Mueni, N. K. (2022). Teachers' pedagogical preparedness for the implementation of the competency-based curriculum in public secondary schools in Kirinyaga County, Kenya. *IOSR Journal of Research & Method in Education*, 12(1), 33–40. <https://doi.org/10.9790/7388-1201033340>
- Murithi, J., & Yoo, J. E. (2021). Teachers' use of ICT in implementing the competency-based curriculum in Kenyan public primary schools. *Innovation and Education*, 3, Article 5. <https://doi.org/10.1186/s42862-021-00012-0>
- Mutai, E. C., Chebet, E., & Oranga, J. (2024). Competency based curriculum implementation and teacher preparedness in public schools in Moiben Sub-County, Uasin Gishu County, Kenya. *Journal of Research Innovation and Implications in Education*, 8(3), 356–365. <https://doi.org/10.59765/kzxwy84>
- Mutua, M. K., & Waweru, S. N. (2023). Teachers' competence and its influence on effective implementation of competency based curriculum in public primary schools in Machakos County, Kenya. *International Journal of Social Science and Human Research*, 6(5). <https://ijsshr.in/v6i5/67.php>
- Mwang'ombe, A. (2021). *Competency Based Curriculum (CBC) in Kenya: Teachers understanding and skills, reality on the ground, successes, challenges and recommendations on the implementation of Competency-Based Curriculum (CBC) in the Kenyan schools*. ResearchGate. <https://www.researchgate.net/publication/356267149>
- Mwangi, N. M., Karuku, S., & Obura, E. (2026). Teacher preparedness for competency-based curriculum in Kenyan schools: Training and perceptions

- prior to implementation. *International Journal of Evaluation and Research in Education*, 15(1), 714–724. <https://doi.org/10.11591/ijere.v15i1.35742>
- Okeyo, S. M., & Mokua, Z. O. (2023). Preparedness of teachers for implementing competency-based curriculum system of education in public secondary schools in Nyamira South Sub-County. *East African Journal of Education Studies*, 6(2), 406–418. <https://doi.org/10.37284/eajes.6.2.1387>
- Ongubo, J. M. (2025). Public primary school teacher preparedness on CBC adoption in Township Ward, Garissa County, Kenya. *International Journal of Research and Innovation in Social Science*, 9(11), 202–211. https://rsisinternational.org/journals/ijriss/uploads/vol9-iss11-pg202-211-202511_pdf.pdf
- Presidential Working Party on Education Reform. (2023). *Report of the Presidential Working Party on Education Reform: Transforming education, training and research for sustainable development in Kenya*. Ministry of Education. <https://www.education.go.ke/sites/default/files/2023-08/B5%20REPORT%20OF%20THE%20PRESIDENTIAL%20WORKING%20PARTY%20ON%20EDUCATION%20REFORM%207th%20JULY%202023%20.pdf>
- Thukia, W. J. (2025). Kenya’s education reform: Embracing a new learning framework. *International Journal of Educational Reform*. Advance online publication. <https://doi.org/10.1177/10567879251331218>
- UNESCO & International Task Force on Teachers for Education 2030. (2024). *Global report on teachers: Addressing teacher shortages and transforming the profession*. UNESCO. <https://teachertaskforce.org/knowledge-hub/global-report-teachers-addressing-teacher-shortages-and-transforming-profession>
- Wawire, B. A., Kiambati, F., Muhia, N., Gathoni, G., & Abuya, B. (2026). Competence-based curriculum implementation in Africa: A scoping review of pedagogical and assessment practices. *Journal of Curriculum Studies*, 58(4). <https://doi.org/10.1080/00220272.2025.2523429>

APPENDICES

The following appendices contain the research instruments and supporting documents for the study.

APPENDIX I: LETTER OF INTRODUCTION

[Full Name of Student]

Mount Kenya University

School of Education

P.O. Box 342-01000

Thika

[Date]

The Head Teacher,

_____ Primary School,

Habaswein Sub-County,

Wajir County.

Dear Sir/Madam,

RE: REQUEST TO COLLECT RESEARCH DATA IN YOUR SCHOOL

I am a postgraduate student at Mount Kenya University pursuing a Master of Education degree in Curriculum Studies. As part of the requirements for the award of the degree, I am carrying out a study titled “Teachers’ Competence and Its Influence on the Effective Implementation of the Competency-Based Curriculum in Public Primary Schools in Habaswein Sub-County, Wajir County, Kenya.”

Your school has been selected to take part in the study. I kindly request your permission to administer questionnaires to some of your teachers, to interview you and to observe one CBC lesson at a time convenient to the school. Participation is voluntary, and the information collected will be treated with strict confidentiality and used for academic purposes only. Neither your name nor the name of your school will appear in any report.

Attached are copies of my introduction letter from the university and the research permit from NACOSTI. Thank you in advance for your cooperation.

Yours faithfully,

[Full Name of Student]

[Registration Number]

APPENDIX II: INFORMED CONSENT FORM

Study title: Teachers' Competence and Its Influence on the Effective Implementation of the Competency-Based Curriculum in Public Primary Schools in Habaswein Sub-County, Wajir County, Kenya.

Purpose of the study: The study seeks to assess teachers' competence in implementing the CBC and to establish how this competence influences the effectiveness of implementation. The findings are expected to inform teacher support programmes and education policy for marginalised areas.

What participation involves: If you agree to take part, you will be asked to complete a questionnaire (about 30 minutes), take part in an interview (about 40 minutes) or allow the researcher to observe one of your lessons, depending on your category of respondent.

Risks and benefits: There are no known risks in taking part in this study. The lesson observation is not an appraisal and will not be shared with your employer. Although you will not receive any payment, your views will contribute to improving teacher support in Habaswein and similar areas.

Confidentiality: Your name will not be recorded on any instrument. All information will be coded, stored securely and used only for this study.

Voluntary participation: Your participation is entirely voluntary. You may decline to answer any question or withdraw at any time without any penalty.

Contact: If you have questions about the study, please contact the researcher on [phone number] or [email address], or the Mount Kenya University Ethics Review Committee.

Declaration by participant

I have read (or have had read to me) the information above. I understand the purpose of the study and what my participation involves, and I voluntarily agree to take part.

Participant's signature: _____ Date: _____

Researcher's signature: _____ Date: _____

APPENDIX III: QUESTIONNAIRE FOR TEACHERS

Instructions

This questionnaire seeks information on teachers' competence and the implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County. Please answer all questions honestly. Do not write your name or the name of your school anywhere on this questionnaire. Tick (✓) the appropriate box or fill in the spaces provided.

Section A: Demographic Information

1. Gender: Male ☐ Female ☐
2. Age bracket (years): Below 30 ☐ 30–39 ☐ 40–49 ☐ 50 and above ☐
3. Highest professional qualification: P1 Certificate ☐ Diploma ☐ Bachelor's degree ☐ Master's degree ☐ Other (specify) _____
4. Employer: TSC ☐ Board of Management ☐ Other (specify) _____
5. Teaching experience (years): Below 5 ☐ 5–10 ☐ 11–15 ☐ Above 15 ☐
6. Grade(s) currently taught: Grade 1–3 ☐ Grade 4–6 ☐ Both ☐
7. Have you attended any CBC training? Yes ☐ No ☐
8. If yes, how many days of CBC training have you attended in total? 1–5 days ☐ 6–10 days ☐ More than 10 days ☐
9. Location of your school: Habaswein town and environs ☐ Rural settlement ☐

Section B: Teachers' Knowledge and Understanding of the CBC

Part I: Indicate the extent to which you agree with each statement. Key: SD = Strongly Disagree (1); D = Disagree (2); N = Neutral (3); A = Agree (4); SA = Strongly Agree (5).

No.	Statement	SD	D	N	A	SA
B1	I clearly understand the vision and mission of the Competency-Based Curriculum.					
B2	I can explain each of the seven core competencies of the CBC.					
B3	I understand how national values should be integrated into lessons.					
B4	I can identify pertinent and contemporary issues (PCIs) relevant to the learning areas I teach.					

No.	Statement	SD	D	N	A	SA
B5	I understand the learning areas and their specific learning outcomes for the grades I teach.					
B6	I understand the KNEC competency-based assessment framework and the four performance levels.					
B7	I understand the role of parents in supporting learning under the CBC.					
B8	I can interpret the curriculum designs for the learning areas I teach without difficulty.					

Part II: For each statement, tick True or False.

No.	Statement	True	False
B9	Under the CBC, the teacher's main role is to act as a facilitator of learning.		
B10	Learner performance in CBC is reported using four levels of achievement.		
B11	Core competencies should only be taught in specific learning areas such as Science and Technology.		
B12	Portfolios, rubrics and observation schedules are examples of CBC assessment tools.		
B13	Summative examinations are the only valid way to assess learners under the CBC.		

Section C: Teachers' Pedagogical Competence

Indicate the extent to which you agree with each statement. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
C1	I prepare schemes of work in the format required for the CBC.					
C2	I prepare lesson plans that clearly state specific learning outcomes and key inquiry questions.					
C3	I regularly use group work, discussion and cooperative learning in my lessons.					
C4	I use inquiry, projects and problem-solving activities to help learners discover knowledge.					
C5	I deliberately integrate core competencies and values into my lessons.					
C6	I improvise teaching and learning resources from locally available materials.					

No.	Statement	SD	D	N	A	SA
C7	I adapt my teaching to meet the needs of learners with different abilities.					
C8	I use a variety of assessment tools such as observation, checklists, rubrics and portfolios.					
C9	I keep up-to-date assessment records for every learner in my class.					
C10	I give learners feedback that helps them to improve their learning.					
C11	I integrate digital or low-cost technology into my lessons where possible.					

Section D: Factors Affecting Teachers' Competence in Implementing the CBC

Indicate the extent to which you agree that each of the following affects your competence in implementing the CBC. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
D1	The CBC training I received was adequate to prepare me to implement the curriculum.					
D2	I have regular opportunities for continuous professional development on the CBC.					
D3	My school has adequate textbooks, curriculum designs and teaching materials.					
D4	My school has adequate ICT facilities, electricity and internet to support CBC teaching.					
D5	My workload allows me enough time to plan and assess learners as required by the CBC.					
D6	The number of learners in my class allows me to use learner-centred methods effectively.					
D7	The head teacher regularly supervises and supports my CBC teaching.					
D8	Curriculum Support Officers visit and support teachers in my school.					
D9	I have a positive attitude towards the CBC.					
D10	Drought and the movement of pastoralist families disrupt my teaching and assessment.					
D11	Insecurity in the area affects my ability to attend training and deliver the curriculum.					
D12	Parents in this community support their children's learning activities under the CBC.					

Section E: Effective Implementation of the CBC

Indicate the extent to which you agree with each statement about CBC implementation in your class. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
E1	Learners actively participate in activities during my lessons.					
E2	Learners in my class demonstrate core competencies such as communication and collaboration.					
E3	PCIs and values are integrated in most of my lessons.					
E4	I use a variety of formative assessment tools as required by the CBC.					
E5	I cover the learning areas in the time allocated in the curriculum designs.					
E6	Learners complete projects and practical activities as required.					
E7	Parents participate in their children's learning activities when asked.					
E8	Overall, the CBC is being implemented effectively in my class.					

Section F: General Comments

1. In your opinion, what are the main challenges you face in implementing the CBC in this sub-county?

2. What kind of support would most improve your competence in implementing the CBC?

3. Any other comments on CBC implementation in Habaswein Sub-County?

Thank you for your time and cooperation.

APPENDIX IV: INTERVIEW GUIDE FOR HEAD TEACHERS AND CURRICULUM SUPPORT OFFICERS

Introduction: Thank you for agreeing to take part in this interview. The purpose is to understand your views on teachers' competence and the implementation of the CBC in Habaswein Sub-County. The interview will take about 40 minutes. With your permission, it will be recorded. Your responses will remain confidential.

Part A: Background

1. How long have you served as a head teacher or Curriculum Support Officer, and how long have you been in Habaswein Sub-County?
2. What CBC training have you personally received?

Part B: Teachers' Knowledge and Understanding of the CBC

3. How would you describe the level of understanding of the CBC among teachers in your school or zone?
4. Which aspects of the curriculum do teachers seem to understand well, and which do they find most difficult?

Part C: Teachers' Pedagogical Competence

5. How do teachers in your school or zone plan and deliver CBC lessons? What methods do they commonly use?
6. How well do teachers carry out competency-based assessment and keep learner records?
7. Can you give an example of a lesson you observed that reflected the CBC very well, or one that did not?

Part D: Factors Affecting Teachers' Competence

8. What training and professional development opportunities are available to teachers here? How adequate are they?
9. How do resources, staffing and class sizes affect teachers' ability to implement the CBC?
10. In what ways do local conditions such as drought, pastoralist mobility and insecurity affect teaching and learning?
11. What support do you provide to teachers, and what challenges do you face in providing it?

Part E: Competence and Effective Implementation

12. In your experience, how does the competence of a teacher affect the way the CBC is implemented in the classroom?
13. What differences do you notice between schools or classes where the CBC is implemented well and those where it is not?
14. What should the Ministry of Education, TSC, KICD and the County Government do to strengthen teachers' competence in this sub-county?

Thank you for your time and valuable contribution.

APPENDIX V: CLASSROOM OBSERVATION CHECKLIST

School code: _____ Grade: _____ Learning area: _____ Date: _____

Number of learners present: _____ Duration of lesson: _____

Rating key: 1 = Not observed; 2 = Partly observed; 3 = Largely observed; 4 = Fully observed

No.	Aspect Observed	1	2	3	4
1	Teacher has a lesson plan in the CBC format with clear learning outcomes				
2	Lesson begins with a key inquiry question or activity that connects to learners' experience				
3	Learners work in pairs or groups during the lesson				
4	Learners are engaged in inquiry, problem solving or hands-on activities				
5	Teacher acts as a facilitator rather than the main source of information				
6	At least one core competency is deliberately developed				
7	Values and/or PCIs are integrated into the lesson				
8	Teacher uses relevant teaching and learning resources, including improvised materials				
9	Technology (digital or low-cost) is used appropriately				
10	Teacher uses formative assessment during the lesson (questions, observation, checklists)				
11	Learners receive feedback on their work				
12	Learners with different abilities are supported				
13	Assessment records and learner portfolios are available and up to date				
14	Lesson concludes with a reflection or summary involving learners				

Observer's field notes:

APPENDIX VI: WORK PLAN

Table A.1: Work Plan, October 2026 – September 2027

Activity	Oct– Nov 2026	Dec 2026– Jan 2027	Feb– Mar 2027	Apr– May 2027	Jun– Jul 2027	Aug– Sep 2027
Proposal correction and defence	✓					
Ethical approval and NACOSTI permit		✓				
Pilot study and revision of instruments		✓				
Main data collection			✓			
Data entry, cleaning and analysis				✓		
Writing of Chapters Four and Five				✓	✓	
Submission of draft thesis to supervisors					✓	
Corrections, final defence and submission						✓

APPENDIX VII: RESEARCH BUDGET

Table A.2: Research Budget

Item	Quantity	Unit Cost (KES)	Total Cost (KES)
Stationery, printing and binding of proposal	6 copies	1,500	9,000
Printing of questionnaires and other instruments	250 sets	60	15,000
NACOSTI research permit	1	1,000	1,000
Training of research assistants	2 persons × 2 days	2,500	10,000
Research assistants' allowance (fieldwork)	2 persons × 30 days	1,500	90,000
Vehicle hire and fuel for fieldwork	30 days	6,000	180,000
Accommodation and meals for researcher	30 days	3,000	90,000
Pilot study expenses	Lump sum	20,000	20,000
Airtime and internet	Lump sum	8,000	8,000
Transcription and translation of interviews	24 interviews	1,000	24,000
Data analysis software and services	Lump sum	25,000	25,000
Printing and binding of final thesis	6 copies	2,000	12,000
Subtotal			484,000
Contingency (10%)			48,400
Grand Total			532,400